

1. Rectangular or Uniform Prob Distribution

Defⁿ: A continuous r.v. X is said to have uniform or rectangular prob distⁿ over the interval (a, b) i.e. $(-\infty < a < b < \infty)$, if its pdf is given by:

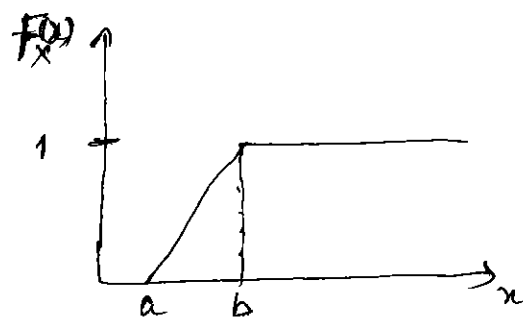
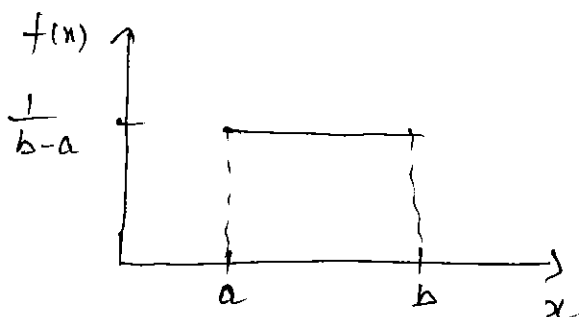
$$f(x; a, b) = \begin{cases} \frac{1}{b-a}, & \text{if } a < x < b \\ 0, & \text{otherwise} \end{cases} \quad \text{--- (*)}$$

Remark: i) $X \sim U(a, b)$

$$\text{ii) } F_X(x) = \int_a^x f(x) dx = \frac{x-a}{b-a}, \quad a < x < b$$

iii) $\frac{d}{dx} F_X(x) = f(x) = \frac{1}{b-a} \neq 0$ exists everywhere except the point $x=a$ and $x=b$ and consequently pdf of x is given by (*)

iv) Graph of $f(x)$ and $F_X(x)$ are given below



v) $X \sim U(-a, a)$

$$f(x) = \frac{1}{2a}, \quad -a < x < a$$

Moments of Rectangular Distⁿ $U(a, b)$

$$\mu_r' = E(X^r) = \int_a^b x^r f(x) dx = \frac{1}{b-a} \int_a^b x^r dx = \frac{1}{b-a} \left(\frac{b^{r+1} - a^{r+1}}{r+1} \right)$$

$$\text{Hence mean } \mu_1' = \frac{1}{b-a} \left(\frac{b^2 - a^2}{2} \right) = \frac{b+a}{2}$$

$$h_2' = \frac{1}{b-a} \left(\frac{b^3 - a^3}{3} \right) = \frac{1}{3} (b^2 + ab + a^2)$$

$$\therefore \text{variance } h_2 = h_2' - h_1'^2 = \frac{1}{3} (b^2 + ab + a^2) - \left(\frac{1}{2} (b+a) \right)^2$$

$$= \frac{1}{12} (b-a)^2$$

MGF of Rectangular (Uniform) distⁿ $U(a, b)$

$$M_X(t) = E e^{tx} = \int_a^b e^{tx} f(x) dx = \int_a^b \frac{e^{tx}}{b-a} dx$$

$$= \frac{e^{bt} - e^{at}}{t(b-a)}, \quad t \neq 0$$

Characteristic Function

$$\phi_X(t) = \int_a^b e^{itx} f(x) dx = \frac{e^{ibt} - e^{iat}}{it(b-a)}, \quad t \neq 0$$

Ex. If X is uniformly distributed with mean 1 and variance $\frac{4}{3}$, find $P(X < 0)$.

If $X \sim U(a, b)$, Mean = $\frac{1}{2} (b+a)$ and

Variance = $\frac{1}{12} (b-a)^2$

Hence $\frac{1}{2} (b+a) = 1$

$\Rightarrow b+a = 2$

$\frac{1}{12} (b-a)^2 = \frac{4}{3}$

$\Rightarrow (b-a)^2 = 16$

Solving we get $a = -1$ and $b = 3$ $\Rightarrow (b-a) = \pm 4$
($a < b$)

$\therefore f(x) = \frac{1}{b-a} = \frac{1}{4}, \quad -1 < x < 3$

Hence $P(X < 0) = \int_{-1}^0 \frac{1}{4} dx = \frac{1}{4} \left[x \right]_{-1}^0 = \frac{1}{4}$

Ex. Subway trains on a certain line run every half hour between midnight and six in the morning. What is the probability that a man entering the station at a random time during this period will have to wait at least 20 minutes?

Solⁿ.

X : denote the waiting time in minutes for the next train.

$$X \sim U(0, 30)$$

$$f(x) = \frac{1}{30}, \quad 0 < x < 30$$

$$\begin{aligned} P(X > 20) &= \int_{20}^{30} f(x) dx = \int_{20}^{30} \frac{1}{30} dx = \frac{1}{30} [x]_{20}^{30} \\ &= \frac{1}{30} [30 - 20] = \frac{10}{30} = \frac{1}{3} \end{aligned}$$

GAMMA DISTRIBUTION

Defⁿ: A r.v. X is said to have a gamma distribution with parameter $\lambda > 0$, if its p.d.f. is given by:

$$f(x) = \begin{cases} \frac{e^{-x} x^{\lambda-1}}{\Gamma(\lambda)} & ; \lambda > 0, 0 < x < \infty \\ 0 & , \text{ otherwise} \end{cases}$$

Remark: (i) $X \sim \sqrt{\lambda}$

$$\begin{aligned} \text{ii) } f(x) \text{ is p.d.f. } \int_0^{\infty} f(x) dx &= \frac{1}{\Gamma(\lambda)} \int_0^{\infty} e^{-x} x^{\lambda-1} dx \\ &= \frac{\Gamma(\lambda)}{\Gamma(\lambda)} = 1. \end{aligned}$$

iii) gamma distⁿ with two parameters.

$$f(x) = \begin{cases} \frac{a^{\lambda}}{\Gamma(\lambda)} e^{-ax} x^{\lambda-1} & ; a > 0, \lambda > 0, \\ 0 & , \text{ otherwise} \end{cases} \quad 0 < x < \infty$$

$$X \sim \sqrt{(a, \lambda)}, \quad X \sim \sqrt{\lambda} = X \sim \sqrt{(1, \lambda)}$$

$$\text{iv) } F_X(x) = P(X \leq x) = \int_0^x f(x) dx = \frac{1}{\Gamma(\lambda)} \int_0^x e^{-u} u^{\lambda-1} du, \quad x > 0$$

MGF of Gamma Distribution $\Gamma(\lambda)$.

$$\begin{aligned}
 M_X(t) &= E e^{tx} = \int_0^{\infty} e^{tx} f(x) dx = \frac{1}{\Gamma(\lambda)} \int_0^{\infty} e^{tx} e^{-x} x^{\lambda-1} dx \\
 &= \frac{1}{\Gamma(\lambda)} \int_0^{\infty} e^{-(1-t)x} x^{\lambda-1} dx = \frac{1}{\Gamma(\lambda)} \cdot \frac{\Gamma(\lambda)}{(1-t)^\lambda}, \quad |t| < 1 \\
 M_X(t) &= (1-t)^{-\lambda}, \quad |t| < 1.
 \end{aligned}$$

Note: For $\Gamma(a, \lambda)$, $M_X(t) = (1 - \frac{t}{a})^{-\lambda}$, $t < a$

$$h_1' = \text{Mean} = \left. \frac{dM_X(t)}{dt} \right|_{t=0}, \quad \frac{\partial M_X(t)}{\partial t} = (-\lambda)(1-t)^{-\lambda-1}$$

$$h_1' = \left. \frac{\partial M_X(t)}{\partial t} \right|_{t=0} = \lambda.$$

$$\begin{aligned}
 h_2' &= \left. \frac{\partial^2 M_X(t)}{\partial t^2} \right|_{t=0}, \quad \frac{\partial^2 M_X(t)}{\partial t^2} = (-\lambda)(-\lambda-1)(1-t)^{-\lambda-2} \\
 &= \lambda(\lambda+1)
 \end{aligned}$$

$$\therefore \text{Variance } h_2 = h_2' - h_1'^2 = \lambda(\lambda+1) - \lambda^2 = \lambda$$

Hence in $\Gamma(\lambda)$, Mean = Variance = λ .

Additive Property : If $X_i \sim \Gamma(\lambda_i)$ ($i=1, 2, \dots, k$)

are independent Γ -variables, then $\sum_{i=1}^k X_i \sim \Gamma\left(\sum_{i=1}^k \lambda_i\right)$

Proof. $M_{X_i}(t) = (1-t)^{-\lambda_i}$

$$M_{X_1+X_2+\dots+X_k}(t) = \prod_{i=1}^k M_{X_i}(t) = (1-t)^{-\sum_{i=1}^k \lambda_i} \text{ which is}$$

the mgf of gamma variate with parameter $\sum_{i=1}^k \lambda_i$.

$$\Rightarrow \sum_{i=1}^k X_i \sim \Gamma\left(\sum_{i=1}^k \lambda_i\right) : \text{ Holds additive property.}$$

Remark : In general if $X_i \sim \Gamma(a, \lambda_i)$, $i=1, 2, \dots, k$ are indep. r.v.s then $\sum_{i=1}^k X_i \sim \Gamma\left(a, \sum_{i=1}^k \lambda_i\right)$.

Beta Distribution of First Kind

Defⁿ: A cont. r.v. x is said to follow a beta distⁿ of first kind with parameters μ and ν ($\mu > 0, \nu > 0$), if its p.d.f. is given by:

$$f(x) = \begin{cases} \frac{1}{\beta(\mu, \nu)} \cdot x^{\mu-1} (1-x)^{\nu-1} & ; (\mu, \nu) > 0, \\ & 0 < x < 1 \\ 0, & \text{otherwise.} \end{cases}$$

where $\beta(\mu, \nu) = \frac{\Gamma(\mu) \Gamma(\nu)}{\Gamma(\mu+\nu)}$, a beta function.

Remark:

i) $X \sim \beta_1(\mu, \nu)$

ii) if $X \sim \beta_1(\mu, \nu)$ then $(1-x) \sim \beta_1(\nu, \mu)$

iii) $F_X(x) = P(X \leq x) = \int_0^x f(x) dx.$

iv) if $\mu=1, \nu=1$ $f(x) = \frac{1}{\beta(1,1)} = 1, 0 < x < 1$

which is $U(0,1)$.

Constants of $\beta_1(\mu, \nu)$

$$\begin{aligned} \mu_1' &= E(X^r) = \int_0^1 x^r f(x) dx = \int_0^1 \frac{1}{\beta(\mu, \nu)} x^{\mu+r-1} (1-x)^{\nu-1} dx \\ &= \frac{1}{\beta(\mu, \nu)} \beta(\mu+r, \nu) = \frac{\Gamma(\mu+r) \Gamma(\nu)}{\Gamma(\mu+r+\nu)} \cdot \frac{\Gamma(\mu+\nu)}{\Gamma(\mu) \Gamma(\nu)} \\ &= \frac{\Gamma(\mu+r) \Gamma(\mu+\nu)}{\Gamma(\mu+r+\nu) \Gamma(\mu)} \end{aligned}$$

In particular $\mu_1' = \frac{\mu}{\mu+\nu}$ (Mean), $\mu_2' = \frac{\mu(\mu+1)}{(\mu+\nu)(\mu+\nu+1)}$

Variance (μ_2) $= \mu_2' - \mu_1'^2 = \frac{\mu\nu}{(\mu+\nu)^2(\mu+\nu+1)}$

Beta distribution of second kind :

Defⁿ: A cont r.v. X is said to have a beta distribution of the second kind with parameters k and v , $(k > 0, v > 0)$, if its p.d.f is given by:

$$f(x) = \begin{cases} \frac{1}{B(k, v)} \cdot \frac{x^{k-1}}{(1+x)^{k+v}} ; & (k, v) > 0, 0 < x < \infty \\ 0, & \text{otherwise.} \end{cases}$$

Remark: 1) $X \sim \beta_2(k, v)$

11) A Beta distⁿ of second kind is transformed to Beta distⁿ of first kind by the transformation: $1+x = \frac{1}{y} \Rightarrow y = \frac{1}{1+x}$

Thus if $X \sim \beta_2(k, v)$, then $Y \sim \beta_1(k, v)$.

Moments

$$\begin{aligned} \mu'_r &= E(X^r) = \int_0^{\infty} x^r f(x) dx = \frac{1}{B(k, v)} \int_0^{\infty} \frac{x^{k+r-1}}{(1+x)^{k+v}} dx \\ &= \frac{1}{B(k, v)} \int_0^{\infty} \frac{x^{(k+r)-1}}{(1+x)^{(k+r)+v-r}} dx = \frac{B(k+r, v-r)}{B(k, v)} \\ &= \frac{\Gamma(k+r) \Gamma(v-r)}{\Gamma(k+v)} \cdot \frac{\Gamma(k+v)}{\Gamma(k) \Gamma(v)} \\ &= \frac{\Gamma(k+r) \Gamma(v-r)}{\Gamma(k) \Gamma(v)} ; \quad v > r \end{aligned}$$

In particular $\mu'_1 = \frac{\Gamma(k+1) \Gamma(v-1)}{\Gamma(k) \Gamma(v)} = \frac{k \Gamma(k) \Gamma(v-1)}{\Gamma(k) (v-1) \Gamma(v-1)} = \frac{k}{v-1}, v > 1$
(Mean)

Similarly $\mu'_2 = \frac{k(k+1)}{(v-1)(v-2)}, v > 2$

$\therefore$ Variance $\mu_2 = \mu'_2 - (\mu'_1)^2 = \frac{k(k+v-1)}{(v-1)^2 (v-2)}, v > 2$

Exponential Distribution :

Defⁿ: A cont. r.v. X is said to follow an exponential distribution with parameter $\theta > 0$, if its p.d.f. is given by :

$$f(x, \theta) = \begin{cases} \theta e^{-\theta x} ; x \geq 0 \\ 0, \text{ otherwise} \end{cases}$$

Remark 1) $X \sim \text{Exp}(\theta)$

$$\begin{aligned} \text{ii) } F_X(x) &= P(X \leq x) = \int_0^x f(x) dx = \int_0^x \theta e^{-\theta x} dx \\ &= 1 - e^{-\theta x}, \quad x \geq 0. \end{aligned}$$

MGF of $\text{Exp}(\theta)$

$$\begin{aligned} M_X(t) &= E e^{tx} = \int_0^{\infty} e^{tx} f(x) dx = \theta \int_0^{\infty} e^{tx} e^{-\theta x} dx \\ &= \theta \int_0^{\infty} e^{-(\theta-t)x} dx = \frac{\theta}{(\theta-t)} = \left(1 - \frac{t}{\theta}\right)^{-1}, \quad \theta > t \end{aligned}$$

$$M_X(t) = \left(1 - \frac{t}{\theta}\right)^{-1}$$

$$h_1' = \text{Mean} = \frac{\partial M_X(t)}{\partial t} \Big|_{t=0}, \quad \frac{\partial M_X(t)}{\partial t} = (-1) \left(1 - \frac{t}{\theta}\right)^{-2} \left(-\frac{1}{\theta}\right)$$

$$h_1' = \frac{\partial M_X(t)}{\partial t} \Big|_{t=0} = \frac{1}{\theta}$$

$$h_2' = \frac{\partial^2 M_X(t)}{\partial t^2} \Big|_{t=0}, \quad \frac{\partial^2 M_X(t)}{\partial t^2} = (-1)(-2) \left(1 - \frac{t}{\theta}\right)^{-3} \left(-\frac{1}{\theta}\right) \left(-\frac{1}{\theta}\right)$$

$$h_2' = \frac{\partial^2 M_X(t)}{\partial t^2} \Big|_{t=0} = \frac{2}{\theta^2}$$

$$\text{Hence Variance } h_2 = h_2' - (h_1')^2 = \frac{2}{\theta^2} - \frac{1}{\theta^2} = \frac{1}{\theta^2}$$

$$\text{Mean} = \frac{1}{\theta}, \quad \text{Variance} = \frac{1}{\theta^2}$$

Remark : Variance = $\frac{1}{\theta^2} = \frac{1}{\theta} \cdot \frac{1}{\theta} = \frac{\text{Mean}}{\theta}$

$\therefore$ Variance $>$ Mean if $0 < \theta < 1$
 Variance $=$ Mean if $\theta = 1$
 Variance $<$ Mean if $\theta > 1$

Hence
 Variance $>, =, <$ Mean
 for different values of the
 parameter.

Ex ~~Proof~~: If $X_1, X_2, \dots, X_n$ are independent r.v.s ~~with~~ follow exponential distribution with parameter $\theta_i; (i=1, 2, \dots, n)$ then $Z = \min(X_1, X_2, \dots, X_n)$ has exponential distⁿ with parameter $\sum_{i=1}^n \theta_i$.

Proof: $G_Z(z) = P(Z \leq z) = 1 - P(Z > z) = 1 - P(\min(X_1, X_2, \dots, X_n) > z)$
 $= 1 - P(X_i > z; i=1, 2, \dots, n) = 1 - \prod_{i=1}^n P(X_i > z)$

($\because X_1, X_2, \dots, X_n$ are independent)

$$= 1 - \prod_{i=1}^n [1 - P(X_i \leq z)] = 1 - \prod_{i=1}^n [1 - F_{X_i}(z)]$$

$$= 1 - \prod_{i=1}^n [1 - (1 - e^{-\theta_i z})] = 1 - e^{-\left(\sum_{i=1}^n \theta_i\right)z}$$

$$\therefore g_Z(z) = \frac{d}{dz} G_Z(z) = \left(\sum_{i=1}^n \theta_i\right) \exp\left(-\sum_{i=1}^n \theta_i\right)z, \quad z > 0$$

Hence $Z = \min(X_1, X_2, \dots, X_n) \sim \text{Exp}\left(\sum_{i=1}^n \theta_i\right)$

Remark: If $X_i; (i=1, 2, \dots, n)$ are iid with parameter θ then $Z = \min(X_1, X_2, \dots, X_n) \sim \text{Exp}(n\theta)$.

Ex. Show that the exponential distribution 'lacks memory' i.e. if $X \sim \text{Exp}(\theta)$, then for every constant $a > 0$, one has $P(Y \leq x | X > a) = P(X \leq x) \forall x$, where $Y = X - a$.

Solⁿ: If $X \sim \text{Exp}(\theta)$, $f(x) = \theta e^{-\theta x}; \theta > 0, 0 < x < \infty$

We have

$$\begin{aligned} P(Y \leq x | X > a) &= P(X - a \leq x | X > a) \\ &= P(X \leq x + a | X > a) = P(a \leq X \leq a + x) \\ &= \theta \int_a^{a+x} e^{-\theta x} dx = \theta \cdot \left(\frac{-1}{\theta}\right) \left[e^{-\theta x} \right]_a^{a+x} = e^{-a\theta} (1 - e^{-\theta x}) \end{aligned}$$

$$\text{and } P(X > a) = \theta \int_a^{\infty} e^{-\theta x} dx = e^{-a\theta}$$

$$\therefore P(Y \leq x | X > a) = \frac{P(Y \leq x \cap X > a)}{P(X > a)} = 1 - e^{-\theta x} \quad \text{--- (*)}$$

$$P(X \leq x) = \theta \int_0^x e^{-\theta x} dx = 1 - e^{-\theta x} \quad \text{--- (10)}$$

Hence from (8) & (10) we get $P(Y \leq x | X > a) = P(X \leq x)$
 i.e. exponential distⁿ lacks memory.

WEIBUL Distribution

A random variable X has a Weibul distⁿ with three parameters $c > 0$, $\alpha > 0$ and k , if the r.v. $Y = \left(\frac{X-k}{\alpha}\right)^c$ has the exponential distⁿ with p.d.f. $f_Y(y) = e^{-y}$, $y > 0$.

Defⁿ: A cont.-r.v. X has a Weibul distⁿ with parameters $c(>0)$, $\alpha(>0)$ and k if its p.d.f. is:

$$f(x; c, \alpha, k) = \frac{c}{\alpha} \left(\frac{x-k}{\alpha}\right)^{c-1} \exp\left\{-\left(\frac{x-k}{\alpha}\right)^c\right\}; x > k, c > 0.$$

The standard Weibul distⁿ ~~which depends~~ is obtained by taking $\alpha=1$ and $k=0$, so that p.d.f. of Weibul distⁿ depends only on a single parameter c is given by

$$f_X(x) = c x^{c-1} \exp(-x^c); x > 0, c > 0.$$

Moments: Moments of Standard Weibul distⁿ:

When $\alpha=1$, $k=0$, $Y = X^c$ which has the exponential distⁿ. We have

$$h_r = E(X^r) = E\left(Y^{\frac{r}{c}}\right) = E Y^{\frac{r}{c}}$$

$$= \int_0^\infty e^{-y} y^{\frac{r}{c}} dy \Rightarrow h_r = \sqrt{\frac{r}{c} + 1}$$

$$\therefore \text{Mean}(h_1) = E(X) = \sqrt{\frac{1}{c} + 1}$$

$$V(X) = h_2 = h_2' - h_1'^2 = \sqrt{\frac{2}{c} + 1} - \left(\sqrt{\frac{1}{c} + 1}\right)^2$$

CAUCHY DISTRIBUTION

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Defⁿ: A r.v. X is said to follow a standard Cauchy distribution if its p.d.f. is given by

$$f_X(x) = \frac{1}{\pi(1+x^2)}; -\infty < x < \infty$$

and X is termed as standard Cauchy variate.

More generally, Cauchy distribution with parameters λ and μ has the p.d.f.:

$$g_Y(y) = \frac{\lambda}{\pi[\lambda^2 + (y-\mu)^2]}; -\infty < y < \infty; \lambda > 0$$

if $Y \sim C(\lambda, \mu)$, then $X = \frac{Y-\mu}{\lambda} \sim C(1, 0)$

1. Characteristic Function of Standard Cauchy Distribution:

$$\phi_X(t) = E e^{itX} = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{e^{itx}}{1+x^2} dx = e^{-|t|}$$

Remarks: If $Y \sim C(\lambda, \mu)$

$$X = \frac{Y-\mu}{\lambda} \sim C(1, 0) \Rightarrow Y = \mu + \lambda X$$

$$\begin{aligned} \therefore \phi_Y(t) &= E e^{itY} = E e^{it(\mu + \lambda X)} = e^{it\mu} \phi_X(t\lambda) \\ &= e^{it\mu} e^{-\lambda|t|} = e^{it\mu - \lambda|t|}; \lambda > 0 \end{aligned}$$

2. Additive Property: If X_1 and X_2 are independent Cauchy variates with parameters (λ_1, μ_1) and (λ_2, μ_2) respectively then $X_1 + X_2$ is a Cauchy variate with parameters $(\lambda_1 + \lambda_2, \mu_1 + \mu_2)$.

$$\phi_{X_j}(t) = e^{it\mu_j - \lambda_j|t|}; j=1, 2$$

$$\phi_{X_1+X_2}(t) = \phi_{X_1}(t) \cdot \phi_{X_2}(t) = e^{it(\mu_1+\mu_2) - (\lambda_1+\lambda_2)|t|}$$

And the ~~the~~ result follows by uniqueness theorem of characteristic functions.

$$Y \sim C(\lambda, 1)$$

3. $\phi_X(t) = e^{i\lambda t - \lambda|t|}; \lambda > 0$

$\frac{\partial \phi_X(t)}{\partial t} = \phi'_X(t)$ does not exist at $t=0$, the mean of the Cauchy distribution does not exist.

In general $\mu, (\sigma, \tau)$ do not exist.

4. Let $X_1, X_2, \dots, X_n$ be a sample of n independent observations from a standard Cauchy distribution and $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, then

$$\begin{aligned} \phi_{\bar{X}}(t) &= \phi_{\sum X_i}(t/n) = \prod_{j=1}^n \phi_{X_j}(t/n) = (\phi_{X_j}(t/n))^n \\ &= (e^{-|t|/n})^n = e^{-|t|} = \phi_X(t). \quad (\text{Since } X_j \text{ are iid}) \end{aligned}$$

Hence by uniqueness theorem of characteristic functions, we have:

"The arithmetic mean $\bar{X}$ of sample $X_1, X_2, \dots, X_n$ of independent observations from a standard Cauchy distribution is also a standard Cauchy variate. In other words, the arithmetic mean of a random sample of any size yields exactly as much information as a single determination of X ."

This implies that the sample mean $\bar{X}$ of random sample of size n , as an estimate of popⁿ mean does not improve with increasing n , which contradicts the Weak Law of Large Numbers (WLLN).

Remark: The role of Cauchy distribution in statistical theory often lies in providing counter examples. e.g. it is often quoted as a distribution for which moments do not exist. It also provides an example to show that $\phi_{X+Y}(t) = \phi_X(t)\phi_Y(t)$ does not imply that X and Y are independent.

Ex. If X have a standard Cauchy distribution. Find the p.d.f. of X^2 and identify its distribution.

Given $f(x) = \frac{1}{\pi} \cdot \frac{1}{(1+x^2)}, -\infty < x < \infty$

The distribution function of $Y = X^2$ say $G_Y(y)$ is

$$\begin{aligned} G_Y(y) &= P(Y \leq y) = P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y}) \\ &= \int_{-\sqrt{y}}^{\sqrt{y}} f(x) dx = 2 \frac{1}{\pi} \int_0^{\sqrt{y}} \frac{dx}{1+x^2} = \frac{2}{\pi} \tan^{-1}(\sqrt{y}), \quad 0 < y < \infty \end{aligned}$$

The p.d.f $g_Y(y)$ of Y is given by

$$\begin{aligned} g_Y(y) &= \frac{d}{dy} [G_Y(y)] = \frac{2}{\pi} \cdot \frac{1}{(1+y)} \cdot \frac{1}{2\sqrt{y}} = \frac{1}{\pi} \frac{y^{-\frac{1}{2}}}{(1+y)} \\ &= \frac{1}{\beta(\frac{1}{2}, \frac{1}{2})} \frac{y^{\frac{1}{2}-1}}{(1+y)^{\frac{1}{2}+\frac{1}{2}}}, \quad y > 0 \end{aligned}$$

This is the p.d.f. of Beta distⁿ of second kind with parameters $(\frac{1}{2}, \frac{1}{2})$.

$$\text{i.e. } X^2 \sim \beta_2\left(\frac{1}{2}, \frac{1}{2}\right).$$

Ex.

If $X \sim N(0,1)$ and $Y \sim N(0,1)$ be independent random variables, then $\frac{X}{Y}$ follows standard Cauchy ~~variate~~ distⁿ.

$$\text{i.e. } Z = \frac{X}{Y}$$

$$f(z) = \frac{1}{\pi} \cdot \frac{1}{(1+z^2)}, \quad -\infty < z < \infty.$$